EMILY DOMANICO

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Collaborative data scientist interested in developing workflows to support effective use of data and geographic information. Background in transportation data analysis, cartography, and data visualization.

WORK EXPERIENCE

Senior Data Scientist, Central Transportation Planning Staff (BRMPO) Boston, MA | Mar 2020 – present

Develop datasets, analytical workflows, interactives, and decision support tools to assist in regional transportation planning. Design and automate cross-cutting data management solutions. Manage personal and team technical projects including budgeting, reporting, and client communication.

- Administer and coordinate agency strategy for data tools and resources including GitHub, Posit/ShinyApps, BigQuery, ArcGIS, Outline (documentation wiki).
- Developed workflows to create and integrate vector tiles into data visualization across project and program work (tippecanoe, pmtiles, AWS).
- Designed and developed coded workflows to create and maintain a database of roadway travel time data (BigQuery, R, INRIX, RITIS) and vehicle crash data (IMPACT, BigQuery, arcgis, GitHub Actions).
- Lead technical staff in the design and development of an interactive app to report on transportation equity trends for the MPO's Transportation Equity Program (Interactive App) (shinyapps, R, pmtiles, Conveyal).
- Designed destination access metrics to assist in regional transportation planning using dasymetric population mapping, R, python, and Conveyal. (https://github.com/CTPSSTAFF/existing-inequities)
- Projected equity implications for fare changes using survey, ridership, and revenue data for the MBTA. Transitioned work to coded process to accommodate more complex fare changes and rolling data input updates. (2024 Fare Changes and 2022 Fare Changes)
- Reported on equity impacts of price changes in the Bluebikes bikeshare system, including estimating the impacts of introducing e-bikes.
- Researched and designed models to estimate rail replacement shuttle passenger counts for the MBTA using R and postgres.
- Researched and adapted algorithms to match bus segments to the road network to support transit priority planning (R, python).
- Staff generated research projects: US Census to transit passenger survey investigation (Interactive App and Report) & Identifying durable bus ridership corridor since the pandemic (Interactive App and Presentation).

Graduate Assistant, Penn State University State College, PA | Aug 2017 – May 2019

Teaching assistant for geography classes. Conducted GIS and cartography labs and developed curricula. Conducted user studies of geographic information visualization strategies to analyze public comments for the Bureau of Land Management. Supporting Women in Geography (SWIG) Officer. Undergrad research mentor.

Mapping Technician, Cruise Automation San Francisco, CA | Mar 2017 – Jul 2017

Developed maps for autonomous cars. Digitized street maps and performed quality assurance. Created and maintained maps for driverless cars using SQL, postGIS, QGIS, and JIRA.

GIS Field Technician, Evari GIS Consulting San Diego, CA | Oct 2016 – Jan 2017

Created maps using GIS software for a state-wide streetlight audit for a utility company. Compiled geographic data from field observation, existing maps, and satellite imagery.

EDUCATION

Penn State University, MS Geography – Cartography and GIS Focus 2017 – 2019

- Thesis: Map Use and Geographic Information Flows in Rail Transit Control Centers

University of California, Berkeley, BA History of Art – Chemistry Minor 2012 – 2016

SKILLS

R, tidyverse, sf, sfnetworks, stars, tidycensus, ShinyApps – python, geopandas, osmnx, networkx, arcgis – postgreSQL, postGIS, SQL, BigQuery – GitHub and git – QGIS, GDAL, ArcGIS Pro – html, css, javascript – Transit and Mobility Data: GTFS, Conveyal, Streetlight, Replica